

LEVEL B/C · VOCABULARY + TASKS

Why Pilots Are Worried About Plans to Replace Co-Pilots with AI

Vocabulary builder, comprehension tasks, and answer key · New Scientist

The words and phrases below appear in the article *Why Pilots Are Worried About Plans to Replace Co-Pilots with AI* by Paul Marks, published in *New Scientist*. The vocabulary spans four registers: aviation operations, safety and risk, regulation and economics, and journalistic expression. Study the table carefully, then complete all three tasks.

PART 1 – VOCABULARY TABLE [reference – no marks]

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
Single Pilot Operations (SPO)	a proposed system in which a commercial airliner is flown by one human pilot supported by AI, rather than two human pilots	Critics argue that Single Pilot Operations would eliminate vital human-to-human interactions on the flight deck.
extended Minimum Crew Operations (eMCO)	a transitional system with two human pilots on board, but with extended periods where only one is at the controls while the other rests	eMCO is planned as a stepping stone before full Single Pilot Operations are introduced.
flight deck	the cockpit of a commercial aircraft — the area where the pilots sit and control the plane	Under SPO, the flight deck would have just one human pilot alongside an AI co-pilot.
co-pilot	the second pilot on a commercial flight, who assists the captain and can take over control if necessary	SPO proposals would replace the human co-pilot with an artificial intelligence system.
sleep inertia	the state of grogginess and reduced alertness that follows waking from sleep, lasting up to 30 minutes	After a rest break, sleep inertia means a pilot needs up to half an hour to regain full alertness.
pilot fatigue	the physical and mental exhaustion experienced by pilots, which can significantly impair judgement and reaction times	The union president argued that SPO would worsen pilot fatigue rather than reduce it.
startle response	the involuntary physical and cognitive reaction triggered by a sudden unexpected event, which can temporarily impair a pilot's ability to respond	Human co-pilots can help calm each other down after a startle response, something AI cannot replicate.

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
gate-to-gate	describing the entire duration of a flight from departure gate to arrival gate, including all phases of the journey	<i>Once eMCO is perfected, SPO could begin fully automated single-pilot gate-to-gate flights.</i>
safety-critical	describes a task, system, or role in which failure could directly lead to loss of life or catastrophic consequences	<i>The role of co-pilot is safety-critical — mistakes in that seat can cost hundreds of lives.</i>
talismanic warning	an event treated as a powerful symbolic lesson or cautionary example that should not be forgotten	<i>The QF32 incident is cited as a talismanic warning against over-reliance on automated systems.</i>
calamity	a sudden, disastrous event causing great damage or suffering	<i>Had the flight computer been flying QF32 alone, calamity could have ensued.</i>
sensor data	information collected by instruments measuring physical conditions such as speed, altitude, fuel levels, and engine status	<i>The QF32 computer was starved of accurate sensor data after the engine explosion severed hundreds of cables.</i>
debilitating	causing severe weakness or inability to function properly; making normal operations extremely difficult	<i>The pilot shortage has become economically debilitating for the aviation industry.</i>
stiff resistance	strong, determined opposition that is difficult to overcome	<i>SPO is meeting stiff resistance from pilot trade unions around the world.</i>
slate	to strongly criticise or condemn something	<i>Unions have slated SPO as nothing more than a cost-cutting measure dressed up as innovation.</i>
backers	people or organisations that support and promote a particular plan or initiative	<i>SPO's backers plan a gradual two-stage build-up to full automation.</i>
regulatory	relating to the rules and standards set by official bodies that control an industry	<i>EASA pushed back any regulatory orders on SPO until after 2027.</i>
quash	to completely suppress or put an end to something	<i>Critics argue that AI co-pilots would quash a host of important human-to-human interactions.</i>
stalking	here: moving quietly and threateningly through a place or field, suggesting a danger that is not yet fully visible	<i>The writer says a dangerous idea is 'stalking' the world of passenger aviation — approaching but not yet arrived.</i>
at the helm	in control or command of something; originally nautical, referring to steering a ship	<i>The proposal would halve the number of human pilots at the helm of civilian airliners.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
furore	a state of great public anger, excitement, or controversy about something	<i>The furore over SPO led EASA to delay any regulatory decisions until after 2027.</i>
cited	referred to as an example or piece of evidence in support of an argument	<i>The QF32 incident was cited by many speakers at the RAeS conference as a warning against SPO.</i>

PART 2 – YES / NO / NOT GIVEN [7 marks] Questions 1–7

Do the following statements agree with the **claims of the writer** in the article?

Write your answers in the boxes below.

YES	<i>if the statement agrees with the claims of the writer</i>
NO	<i>if the statement contradicts the claims of the writer</i>
NOT GIVEN	<i>if it is impossible to say what the writer thinks about this</i>

1	The writer believes that replacing human co-pilots with AI is a good idea. Answer: _____
2	Airlines argue that SPO will help solve the problem of a shortage of qualified pilots. Answer: _____
3	An AI co-pilot would be able to replicate the calming effect that human pilots have on each other during emergencies. Answer: _____
4	The QF32 incident shows that aircraft computers can always be trusted to make safe decisions with incomplete data. Answer: _____
5	EASA has permanently cancelled all plans to introduce SPO on commercial flights. Answer: _____
6	The writer suggests that ordinary passengers have some power to influence whether SPO is introduced. Answer: _____
7	Air transport economist Floris de Haan believes SPO will definitely go ahead regardless of public opinion. Answer: _____

PART 3 – GAP-FILL [16 marks] Questions 8-15

Complete the sentences using the correct word or phrase from the word bank below. You may need to change the form of the word. Most sentences require more than one answer.

WORD BANK

Single Pilot Operations (SPO)	extended Minimum Crew Operations (eMCO)	flight deck	co-pilot
sleep inertia	pilot fatigue	startle response	gate-to-gate
safety-critical	talismanic warning	calamity	sensor data
debilitating	stiff resistance	slate	backers
regulatory	quash	stalking	at the helm
furore	cited		

1	Under _____, a commercial flight would have just one human pilot on the _____ alongside an AI designed to carry out the _____ role of _____.
2	The airline industry has faced an economically _____ pilot shortage, which _____ of SPO argue could be solved by replacing the human _____ with AI.
3	Critics insist that an AI system cannot replicate the way two human pilots manage each other's _____ response, cover for each other during rest breaks when _____ affects alertness, or address the risk of _____.
4	The proposed two-stage approach would first introduce _____, in which two pilots are on board but one rests while the other manages the controls with AI support, before moving to full _____ flights.
5	The QF32 incident is treated as a _____ warning: when an engine explosion left the flight computer _____ of accurate _____, it instructed pilots to transfer fuel into a damaged tank — a decision that, if followed, could have led to _____.
6	The _____ over SPO has caused EASA to delay any _____ orders — the agency was already meeting _____ from pilot unions who _____ the proposal as cost-cutting.
7	The writer opens with a vivid metaphor, describing a dangerous idea as _____ the world of passenger aviation — and ends with advice to write to airlines to make consumer views known, given that 'the power of the consumer' will ultimately decide whether SPO proceeds.
8	SPO's _____ argue the technology will eventually improve safety, but unions and researchers suggest the real risk lies in removing the human interactions that keep pilots _____ and able to respond when AI systems are _____ of the data they need to function correctly.

Write your answers here:

1. _____	2. _____
3. _____	4. _____
5. _____	6. _____
7. _____	8. _____

ANSWER KEY

Answer Key — Why Pilots Are Worried About Plans to Replace Co-Pilots with AI

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Part	Task	Qs	Marks
1	Vocabulary Table	—	Reference only
2	Yes / No / Not Given	1–7	7
3	Gap-Fill (1 mark per blank)	8–15	16 (approx)
TOTAL			~23 marks

Part 2 — YES / NO / NOT GIVEN [7 marks]

These test the writer's CLAIMS. Accept YES / NO / NOT GIVEN or Y / N / NG.

- | | |
|---------------------|---|
| 1. NO | <i>The writer explicitly calls SPO 'a dangerous idea' and argues it 'could make flying far less safe'.</i> |
| 2. YES | <i>Para 2: 'This, airlines argue, will address a pilot shortage that has become economically debilitating.'</i> |
| 3. NO | <i>Para 3: 'it won't replicate how pilots can calm each other in stressful situations.'</i> |
| 4. NO | <i>The QF32 example shows the opposite — the computer gave a dangerous instruction when starved of data.</i> |
| 5. NOT GIVEN | <i>EASA pushed back regulatory orders until after 2027 — the article does not say plans are permanently cancelled.</i> |
| 6. YES | <i>Final para: 'The power of the consumer will decide if SPO goes ahead' — writer advises readers to write to airlines.</i> |
| 7. NO | <i>De Haan said 'the power of the consumer will decide' — he explicitly gave power to public opinion, not fate.</i> |

Part 3 — Gap-Fill [1 mark per blank]

Award 1 mark per blank filled correctly. Accept minor spelling variations if clearly the correct word.

- | | |
|--|--------------------------------|
| 8. Single Pilot Operations (SPO) / flight deck / safety-critical / co-pilot | <i>4 blanks — 1 mark each.</i> |
| 9. debilitating / backers / co-pilot | <i>3 blanks — 1 mark each.</i> |
| 10. startle (response) / sleep inertia / pilot fatigue | <i>3 blanks — 1 mark each.</i> |

11. **extended Minimum Crew Operations (eMCO) / gate-to-gate** 2 blanks — 1 mark each.
12. **talismanic / starved / sensor data / calamity** 4 blanks — 1 mark each.
13. **furore / regulatory / stiff resistance / slated** 4 blanks — 1 mark each.
14. **stalking** 1 blank.
15. **backers / at the helm / starved** 3 blanks — 1 mark each. Accept: quash for third blank in context.

MARKING SUMMARY

Part 2 (Y/N/NG): 7 marks — 1 per question · Part 3 (Gap-fill): approx. 16 marks — 1 per blank · TOTAL: approx. 23 marks · Part 1 is reference only — no marks awarded

ANSWER KEY · WHY PILOTS ARE WORRIED — AI CO-PILOTS

KEY